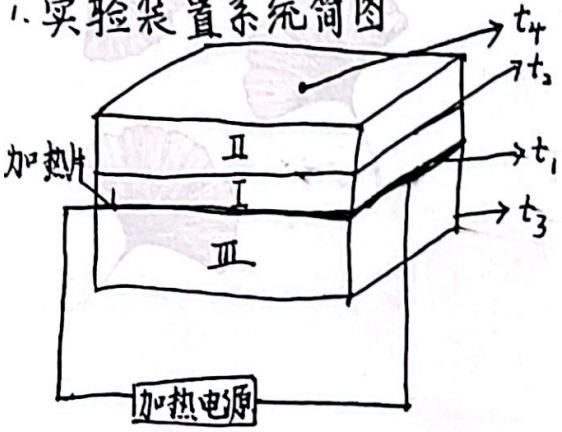
班级: 姓名: 学号: 科目: 第 页

常功率平面热源法同时测定绝热材料导热系数λ和导温系数a

1. 实验装置系统简图



2. $\tau = 600s$ 时 $T_1 = 27.5°C$ $T_2 = 25.01°C$

$ierfc\left(\frac{x_1}{2\sqrt{a\tau}}\right) = \frac{\phi}{\sqrt{\pi}}$ $\therefore \frac{x_1}{2\sqrt{a\tau}} = 0.74$ (1.08, 0.74)

$a = 0.74 \times 10^{-6} m^2/s$ (1.61, $0.459 \times 10^{-7} m^2/s$)

$\lambda = \frac{2\phi_0}{\theta_{0,\tau}}\sqrt{a\tau}\frac{1}{\sqrt{\pi}} = 0.137 W/(m \cdot K)$

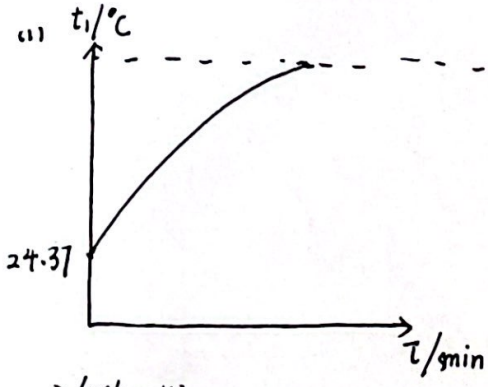
$\tau = 7205s$ 时 $T_1 = 27.89°C$ $T_2 = 25.22°C$

$ierfc\left(\frac{x_1}{2\sqrt{a\tau}}\right) = \frac{\phi}{\sqrt{\pi}}$ $\therefore \frac{x_1}{2\sqrt{a\tau}} = 0.66$ (1.20, 0.66)

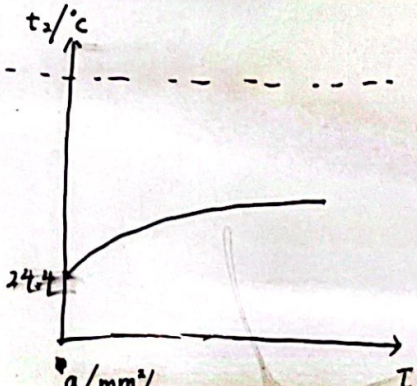
$a = 0.81 \times 10^{-7} m^2/s$ (1.69)

$\lambda = \frac{2\phi_0}{\theta_{0,\tau}}\sqrt{a\tau}\frac{1}{\sqrt{\pi}} = 0.146 W/(m \cdot K)$

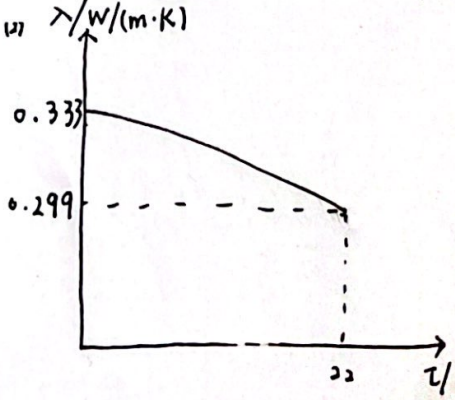
3. (1) $t_1/°C$



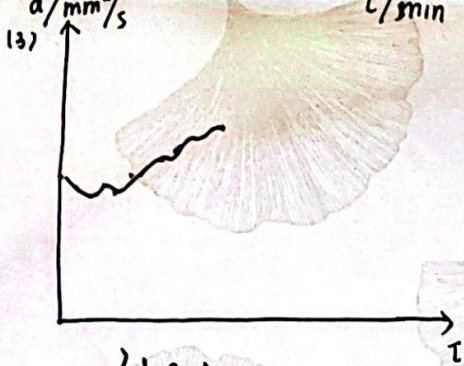
$t_2/°C$



(2) $\lambda/W/(m \cdot K)$



(3) $a/mm^2/s$

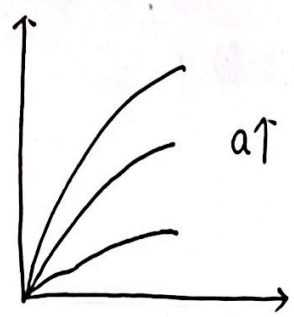


固体随温度升高，晶格振动加剧阻碍热量传输导热系数下降，且材料本身不均匀

过余温度小，相对误差大，材料不均匀

4. $x_1 = 14.56\,mm$ $\quad ierfc\left(\frac{x_1}{\sqrt{a\tau}}\right) = \frac{\phi}{\sqrt{\pi}}$ $\quad \lambda = \frac{2q_0}{\theta_{0,\tau}}\sqrt{a\tau}\frac{1}{\sqrt{\pi}}$

(1) 固定 λ 改变 a

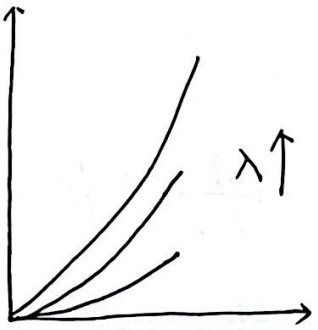


a 越大，$\frac{x_1}{\sqrt{a\tau}}$ 越小，$ierfc\left(\frac{x_1}{\sqrt{a\tau}}\right)$ 越大，$\phi = \frac{\theta_{x_1,\tau}}{\theta_{0,\tau}}$ 越大

且 λ 固定，$\theta_{0\tau}$ 越大，所以 $\theta_{x_1,\tau}$ 越大，

t_2 越大

(2) 固定 a 改变 λ



a 固定 则 $\phi = \frac{\theta_{x_1,\tau}}{\theta_{0,\tau}}$ 固定，λ 越大，$\theta_{0,\tau}$ 越小，

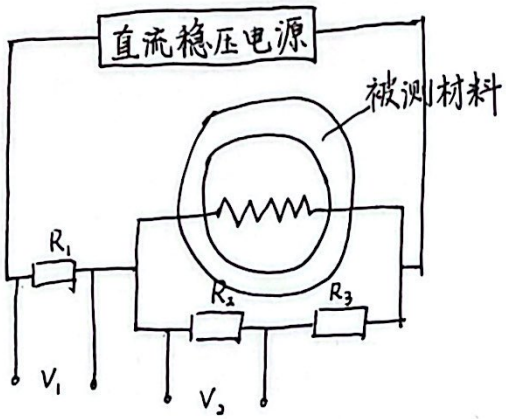
则 $\theta_{x_1,\tau}$ 越大，所以 t_2 越大

作业纸

班级： 姓名： 学号： 科目： 第 页

圆球法测定绝热材料导热系数

1.



2. $d_1 = 100\,mm$ $d_2 = 150\,mm$

	内球平均温度°C	外球平均温度°C	功率 W	导热系数 W/(m·K)
木炭 (6V)	42.16	31.61	1.838	0.0924
木炭 (10V)	67.29	40.79	5.370	0.1074
小米 (10V)	59.72	40.91	5.385	0.1519

3. $d_1 = 100\,mm$ $d_2 = 150\,mm$

	t_1/°C	t_2/°C	$\bar{t}$/°C	Q/W	λ/W/(m·K)	λ($\bar{t}$)/W/(m·K)
木炭	42.16	31.61	36.89	1.838	0.0924	0.0924
	67.29	40.79	54.04	5.370	0.1074	0.1075
小米	59.72	40.91	50.32	5.385	0.1519	0.1519

4. (1) t_1, t_2 测温精度不够，可换用更高精度的仪表

(2) 被测材料不均匀，可将材料尽量填满被测层并压实排除空隙

5、球坐标系下，Laplace 算子为 $\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$

$\therefore \rho c\frac{\partial t}{\partial\tau} = q_v + \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\lambda\frac{\partial t}{\partial r}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\lambda\sin\theta\frac{\partial t}{\partial\theta}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial\varphi}\left(\lambda\frac{\partial t}{\partial\varphi}\right)$

求解 $\frac{\partial}{\partial\theta} = 0$ $\frac{\partial}{\partial\varphi} = 0$ $\frac{\partial}{\partial\tau} = 0$ $q_v = 0$

$\therefore \frac{\partial}{\partial r}\left(r^2\frac{\partial t}{\partial r}\right) = 0$ 边界条件 $r = r_1$ $t = t_1$; $r = r_2$ $t = t_2$

$\therefore t = -\frac{r_1 r_2}{r_1 - r_2}(t_1 - t_2)\frac{1}{r} + \frac{r_1 t_1 - r_2 t_2}{r_1 - r_2}$

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6. $d_1 = 0.1m$ $d_2 = 0.3m$

	$t_1/°C$	$t_2/°C$	$\lambda/W/(m\cdot K)$	Q/W
木	370	30	0.0924	29.61
泡	370	30	0.1075	34.45
小米	370	30	0.1579	50.60

$$Q = \frac{2\pi\lambda d_1 d_2 (t_1 - t_2)}{d_2 - d_1}$$